ICAI STANDARDS & STATUTORY DIRECT/INDIRECT TAX ENGINEERING

Product Specification: Autonomous Practice Platform for Indian CA Firms

An Enterprise Compliance Operating System designed for Indian Chartered Accountancy practices — embedding 27 bounded agents, 18 human gates, 4 foundational operating laws, and a zero-LLM deterministic mathematical core.

Target Audience: Managing Partners, Tax & Audit Partners, Peer Reviewers | **Statutory Baseline:** CGST/SGST Acts (w/ 2024-26 IMS), IT Act 2025, ICAI SQC 1

Core Invariant: "The Machine Prepares. The Professional Decides." | **Architecture:** 3-Stratum Division of Labour

27	18	143	0.00%
BOUNDED AGENTS	HUMAN SIGN-OFF GATES	STATUTORY DATES / YEAR	LLM RETURN CALCULATION

1.0 THE EXECUTIVE MANDATE: SOLVING THE CA PRACTICE REALITY

PRACTICE REALITY

An Indian CA firm handling 150 to 500 corporate/SME clients navigates over **143 statutory compliance dates annually** across GST, Direct Tax, TDS, MCA, and Payroll. Today, **over 72% of billable professional time is consumed by low-leverage clerical tasks**: chasing clients for missing bank statements and purchase bills over WhatsApp, manually keying in ledgers into Tally, three-way reconciling purchase registers against GSTR-2B and the newly introduced **Invoice Management System (IMS)**, and checking Section 194C/J TDS rates.

Meanwhile, the personal statutory liability of the signing Partner has reached an all-time peak under **ICAI UDIN mandates**, **Class 3 DSC token accountability**, and aggressive GST penal provisions (Section 122/132). **KARYA does not attempt to replace the Chartered Accountant.** It acts as an institutional-grade compiler and reconciler: automating extraction, triage, matching, and preliminary drafting, while forcing all tax judgments, filing submissions, and DSC applications through qualified human gates.

2.0 THE FOUR FOUNDATIONAL OPERATING LAWS (ICAI COMPLIANCE SHIELD)

ETHICAL & LEGAL INVARIANTS

LAW 1: The Machine Prepares. A Person Decides.

ZERO AUTONOMOUS FILING

Across all 27 software agents in the platform, **exactly zero agents hold the authority or physical tools to sign, certify, or submit returns**. Authority to file does not exist in any agent manifest and cannot be unlocked via prompt or configuration. Filing requires explicit Class 3 DSC PIN entry or OTP validation by a licensed human.

LAW 2: Arithmetic is Code, Not Language.

ZERO LLM MATH

Large Language Models (LLMs) are strictly banned from computing return figures. All dates, interest calculations (**Section 50 net cash basis**), late fees (Section 47), slab rates, and 38 statutory blocking checks run inside a deterministic, compiled Python/Rust rules engine. LLMs are confined strictly to OCR translation, document classification, and narrative drafting.

LAW 3: Permissions are Tools, Not Instructions.

AST HARD-BLOCK

Agent behavior is not constrained by "system prompts" or conversational promises. Compliance is enforced at the Abstract Syntax Tree (AST) code level. An agent that extracts purchase bills is physically instantiated without any GSTN API gateway or filing SDK in its execution container. It is mathematically incapable of unauthorized actions.

LAW 4: The Client Proposes. Only the Firm Commits.

PARTNER CONTROL

When an SME client marks an inward invoice on the portal or WhatsApp (e.g. marking "Accept" on an IMS invoice), the action is classified strictly as a *Proposal*. No client proposal ever satisfies a statutory review requirement or commits to government portals until an authorized firm reviewer signs off at Gate G3.

3.0 THE THREE-STRATUM DIVISION OF LABOUR

SYSTEM TOPOLOGY

STRATUM LAYER	TECHNOLOGY & ENGINE	OPERATIONAL SCOPE & RESPONSIBILITIES
Stratum 1: Deterministic Core The Rule & Math Foundation	RUST / PYTHON CORE Zero-LLM Runtime	• **Statutory Calendar:** Generates and tracks 143 client compliance deadlines. • **Deterministic Tax Math:** Section 50 interest on cash ledger, Section 47 late fees, tax slabs. • **38 Rule Checks:** Hard-coded blocking checks (DRC-01B/01C mismatch, blocked ITC 17(5)). • **Payloads & FVU:** Validated JSON generation for GSTR-1, 1A, 3B, and NSDL TDS FVU bridge.
Stratum 2: Bounded Agent Layer Cognitive Assistants	DUAL ENGINE System 1 (Jev) + System 2 (LLM)	• **Intake (IN-1 to 4):** VLM/OCR bill parsing, bank statement regex parsing, COA ledger mapping. • **GST Line (GS-1 to 6):** IMS invoice triage, sales 3-way reco, dynamic 2B input credit, supplier recovery. • **Direct Tax (DT-1 to 4):** TDS section categorization (194C vs 194J), AIS/26AS reconciliation. • **Firm Ops (FO-1 to 5):** Exception routing, Section 141(3)(g) 60-audit ceiling counter, regulatory watch.
Stratum 3: Human Gate Tier Legal Attestation & DSC	18 HUMAN GATES 10 Staff (L2) + 8 Partner (L3)	• **Tier L2 Staff Review:** Low-confidence OCR review (G2), IMS commit (G3), manual journal review (G5). • **Four-Eyes Verification (G6):** Independent secondary review before any return payload is locked. • **Tier L3 Licensed Partner Gates:** Class 3 DSC token signing (G7), ICAI UDIN generation (G8), Tax Notice reply approval (G11), Annual Tax Audit Form 3CD / GSTR-9C statutory attestation (G15).

The CA Sovereign Data Guarantee: All client databases, financial ledgers, and documents reside within Tier-4 data centers in the Republic of India (AWS Mumbai / Azure Pune). Zero client financial records are retained or utilized to train external AI models. Digital Signature (DSC) cryptographic PINs are held strictly in local ephemeral volatile RAM for <30 seconds and never stored on disk.

4.0 THE COMPLIANCE ASSEMBLY LINE: INTAKE & GST LINE OF SERVICE

DETERMINISTIC & AGENT DIVISION

The platform structures all firm operations into an industrial-grade compliance assembly line. Every machine agent acts as an exhaustive clerk: it fetches, normalizes, cross-references, and drafts. **Two foundational components were demoted to deterministic code:** (1) The Compliance Calendar Engine (143 fixed dates) and (2) The UDIN Generation Bridge, because deterministic rules require 100% auditability with zero probabilistic variance.

> **Architectural Invariant:** Every agent below possesses strictly read and draft tools. No agent possesses tools to hit government submission endpoints, alter finalized client ledgers without human review, or bypass gate thresholds.

FAMILY A: DOCUMENT INTAKE, MULTIMODAL OCR & LEDGER EXTRACTION

ID	AGENT NAME	RUNTIME ENGINE	OPERATIONAL MANDATE	AUTHORITY BOUNDARY & HUMAN GATE
IN-1	Document Intake & OCR	Vision-Language (VLM)	Extracts line items, GSTINs, HSNs, taxable values from scanned invoices, mobile photos, and PDFs. Deduplicates files.	**Gate G2:** Confidence <95% halts to human bounding-box review. Banned from modifying vendor ledger.
IN-2	Client Document Chase	WhatsApp Bot (WABA)	Executes polite 3-day / 1-day / overdue progressive chase ladder on WhatsApp for missing bank statements and bills.	**Gate G1:** Partner can pause or stop chasing. Banned from answering legal or tax queries.
IN-3	COA & Ledger Classifier	TypeSafe Jev Model	Maps transactions to client's specific Tally/Busy Chart of Accounts (COA) based on 3-year historical firm memory.	**Gate G5:** Manual journal adjustments >₹50,000 require secondary four-eyes staff sign-off.
IN-4	Bank Statement Parser	Regex / Narration NLP	Parses statements across 30+ Indian banks. Decodes messy narration (NEFT/RTGS/UPI) to detect counterparty GSTINs.	Flags un-reconciled contra entries. Cannot balance ledger without human review.

FAMILY B: GST INDIRECT TAX COMPLIANCE & RECONCILIATION

ID	AGENT NAME	RUNTIME ENGINE	OPERATIONAL MANDATE	AUTHORITY BOUNDARY & HUMAN GATE
GS-1	Sales Reconciliation	Deterministic Math	Four-way reconciliation: Sales Register (Books) vs GSTR-1 draft vs E-Way Bill Portal vs E-Invoice IRN repository.	Generates variance matrix. Zero authority to edit outward supply figures.
GS-2	IMS Invoice Triage	TypeSafe Jev Engine	Evaluates inward invoices on the new Invoice Management System. Recommends: Accept, Reject, or Keep Pending.	**Gate G3:** Agent only proposes; Licensed reviewer must commit IMS actions to GSTN.
GS-3	Dynamic ITC Matcher	Deterministic SQL Engine	Reconciles GSTR-2B vs Purchase Register. Enforces Section 16(2)(aa), Section 17(5) blocked credits, Rule 37 (180 days).	**Gate G4:** Mismatches >₹50,000 route directly to Partner review.
GS-4	Supplier Recovery Notice	Sovereign Legal LLM	Drafts formal recovery notices to defaulting vendors whose non-filing of GSTR-1 blocks the client's input tax credit.	Drafts communications only; must be approved by client/staff before dispatch.
GS-5	GSTR-3B Return Drafter	Python Rules Engine	Computes net cash liability, auto-utilizes electronic cash/credit ledgers, runs 38 statutory blocking checks.	**Gate G6:** Four-eyes review mandatory. Zero autonomous filing capability.
GS-6	Annual GSTR-9 / 9C Engine	Audit Reconciler	Annual cross-period reconciliation (Table 4, 5, 6, 7, 8, 9). Auto-computes spillover ITC and spillover turnover.	**Gate G15:** Mandatory Partner statutory certification. Cannot file without UDIN.

Why the Calendar Keeper & UDIN Clerk Were Demoted from Agents to Code

ENGINEERING RATIONALE

In architectural revisions 1 and 2, the statutory calendar and UDIN generator were conceived as autonomous software agents. In Revision 3, both were permanently demoted to deterministic code. **Statutory due dates have exactly one legally correct answer under the CGST Act and Income Tax Rules.** Modeling them as probabilistic agents introduces non-deterministic hallucination risk while adding zero business value. In KARYA, statutory dates and UDIN pre-fills run as compiled, immutable, dated state machines.

4.1 COMPLETE AGENT CATALOGUE: DIRECT TAX, AUDIT, SECRETARIAL & OPERATIONS

REMAINING 17 AGENTS

FAMILY C: DIRECT TAX, WITHHOLDING (TDS/TCS) & ADVANCE TAX

ID	AGENT NAME	RUNTIME ENGINE	OPERATIONAL MANDATE	AUTHORITY BOUNDARY & HUMAN GATE
DT-1	TDS/TCS Withholding Drafter	Section Classifier	Determines section applicability (194C vs 194J, 194I, 194Q vs 206C(1H)). Validates lower deduction certificates.	Generates 26Q/24Q quarterly draft. Passes data to NSDL FVU utility.
DT-2	AIS / TIS / 26AS Cross-Check	3-Way Reconciler	Cross-checks books against Income Tax portal AIS/TIS/26AS. Identifies un-credited TDS or unreported income.	Flags variance items. Prohibited from adjusting tax credits without supporting challans.
DT-3	Advance Tax Estimator	Tax Slab Engine	Quarterly P&L projection (June 15, Sept 15, Dec 15, March 15). Calculates Section 234B and 234C interest penalties.	**Gate G14:** Tax estimation methodology and disallowance assumptions require Partner sign-off.
DT-4	ITR Pre-Filler	ITR Schema Engine	Pre-fills ITR-1 through 7 schedules: depreciation (Schedule DAA), capital gains, business profits, MAT/AMT.	**Gate G6 & G7:** Requires four-eyes staff audit and Partner Class 3 DSC token signing.

FAMILY D: AUDIT, ATTESTATION, SECRETARIAL & STATUTORY PAYROLL

ID	AGENT NAME	RUNTIME ENGINE	OPERATIONAL MANDATE	AUTHORITY BOUNDARY & HUMAN GATE
AU-1	Audit Working Paper Compiler	SA 230 Standards Engine	Assembles digital audit files (trial balance, confirmations, materiality sampling, substantive testing memos).	Generates standard ICAI documentation. Prohibited from closing audit queries.
AU-2	Tax Audit Form 3CD Drafter	44-Clause Verifier	Auto-extracts and verifies 44 clauses: MSME Section 43B(h), disallowances 40(a)(ia), related party 40A(2)(b).	**Gate G15:** Partner review and UDIN attestation mandatory for all 44 clauses.
CO-1	MCA & ROC Secretarial	ROC Schema Engine	Tracks ROC statutory filings: AOC-4, MGT-7, DIR-3 KYC, MBP-1 disclosures, and statutory registers.	Prepares draft forms. V-3 portal filing requires human DSC login.
PY-1	Statutory Payroll Dues	Wage Code Engine	Computes EPF (ECR file), ESIC challans, and state-wise Professional Tax (PT) strictly on the 15th of each month.	**Gate G17:** Provident fund dues are trust money; requires authorized staff payment release.

FAMILY E: FIRM OPERATIONS, PRACTICE CONTROLS & CLIENT GATEWAY

ID	AGENT NAME	RUNTIME ENGINE	OPERATIONAL MANDATE	AUTHORITY BOUNDARY & HUMAN GATE
FL-1	Portal Filing Watcher	GSTN/ITD Webhook	Monitors government servers post-submission. Captures ARN acknowledgements and securely vaults signed PDFs.	Read-only scraper. Cannot initiate any transaction or submission.
FO-1	Audit Ceiling Guard	Sec 141(3)(g) Tracker	Maintains real-time counter of company audits signed per partner against statutory limit of 60 across all firms.	Hard blocks dispatch of audit engagements exceeding partner ceiling.
FO-2	Exception Triage & Router	Priority Dispatcher	Aggregates escalations from all 26 agents; prioritizes same-day statutory deadlines directly to partner desks.	Sorts and assigns tasks; possesses zero operational decision authority.
FO-3	Billing & Recovery Engine	Practice Ledger Engine	Detects out-of-scope work (notice handling, DRC-01 replies, extra revisions) and generates draft fee bills.	**Gate G9:** Partner must review and approve fee bill before dispatch to client.
FO-4	Regulatory Watch & Impact	Daily Gazette Scraper	Scrapes CBIC, CBDT, MCA notifications; maps statutory changes to specific client industries and flags impacted clients.	Read-only legal repository; cannot modify system rule book without engineering commit.
FO-5	SQC 1 Quality & Peer Review	Adversarial Auditor	Re-evaluates a random 10% sample of completed filings. Injects synthetic canary errors to verify reviewer diligence.	Reports directly to Firm Managing Partner; independent of delivery team.
CL-1	Front Desk (7-Question Guard)	Closed NLP Whitelist	Answers client inquiries on WhatsApp strictly restricted to 7 operational questions (filing status, dues, etc.).	**Gate G10:** Any tax planning, notice interpretation, or legal inquiry is instantly escalated to staff.
CL-2	Tax Notice & Demand Parser	Sovereign Notice OCR	Parses GST DRC-01/01A and IT Sec 143/148 notices; extracts demand amounts, sections, and statutory reply deadlines.	**Gate G11:** Partner exclusively writes and signs statutory notice replies.
KN-1	Legal Reference Shelf	Tax Statute RAG	Answers staff queries regarding GST/Income Tax sections and circulars with statutory citations. Zero client data access.	Reference tool only. Completely air-gapped from client ledgers and PII.

5.0 THE 18 HUMAN-IN-THE-LOOP GOVERNANCE GATES

LEGAL ACCOUNTABILITY STRUCTURE

To satisfy the Institute of Chartered Accountants of India (ICAI) Code of Ethics and Standard on Quality Control (SQC 1), the platform divides all operational sign-offs into **18 non-bypassable Human Gates**. These are split across two tiers: Tier L2 (Staff & Article Assistant Operational Verification) and Tier L3 (Licensed Signing Partner Statutory Commitment).

TIER L2: STAFF & REVIEWER GATES (10 GATES)

OPERATIONAL CONTROL

GATE	VERIFICATION POINT	REVIEWER ROLE & HUMAN ACTION
G1	Client Chase Escalation	Staff calls off or escalates WhatsApp chasing based on client relationship.
G2	Low-Confidence OCR	Side-by-side verification of original invoice image vs extracted figures.
G3	IMS Decision Commit	Staff commits Accept/Reject/Pend proposals to GSTN Invoice Management System.
G4	ITC Mismatch Review	Reviews purchase vs 2B credit gaps (>₹50,000 auto-escalates to Partner).
G5	Manual Journal Entries	Four-eyes verification on non-standard journal entries into Tally/Busy.
G6	Four-Eyes Return Approval	Independent secondary staff reviews complete GSTR/ITR draft before partner sign-off.
G9	Client Fee Note Review	Staff reviews time records and out-of-scope work before bill is finalized.
G10	Non-Standard Client Query	Routes non-standard tax questions from WhatsApp to designated staff desk.
G12	Document Release to Client	Authorized staff approves sending filing acknowledgements to client vault.
G17	Payroll & EPF Release	Authorized staff verifies monthly salary register before EPF/ESIC challan generation.

TIER L3: LICENSED PARTNER GATES (8 GATES)

PARTNER DSC ONLY

GATE	VERIFICATION POINT	STATUTORY REQUIREMENT & PARTNER ACTION
G7	Class 3 DSC Sign-Off	Partner physically enters hardware token PIN into ephemeral RAM to sign returns.
G8	ICAI UDIN Generation	Partner authorizes generation of 18-digit mandatory ICAI Unique Document ID.
G11	Statutory Notice Reply	Partner writes and approves all submissions to GST/IT tax officers.
G13	Client Onboarding (AML)	Partner approves new client engagement, KYC, and conflict-of-interest check.
G14	Tax Position & Disallowance	Partner approves aggressive or debatable tax positions in advance tax/ITR.
G15	Annual Audit Certification	Partner signs Form 3CD Tax Audit and GSTR-9C Annual GST Reconciliation.
G16	Third-Party Data Disclosure	Partner authorizes sharing financial statements with banks or auditors.
G18	Engagement Resignation	Formal documentation and mandatory statutory communication to incoming auditor.

6.0 CLASS 3 DSC HARDWARE SECURITY & UDIN INTEGRATION ARCHITECTURE

ZERO-TRUST CRYPTOGRAPHY

Zero-Trust Ephemeral DSC RAM Architecture

SECURITY PROTOCOL

Chartered Accountants are legally and personally liable for all filings signed under their **Class 3 Digital Signature Certificate (DSC)**. Many legacy software vendors improperly ask CAs to upload PFX certificates or store USB token PINs on cloud servers—a grave breach of Information Technology Act standards.

- **Local USB Cryptoki (PKCS#11) Bridge:** The physical USB ePass2003 / mToken remains plugged into the Partner's local PC.
- **Zero Cloud Pin Storage:** The PIN is entered by the Partner and held strictly in local encrypted volatile RAM for **<30 seconds** during cryptographic payload signing.
- **Signed Digest Transmission:** Only the resulting PKCS#7 cryptographic signature hash is transmitted to the GSTN/ITD gateway. The private key never leaves the physical silicon token.

Automated ICAI UDIN Pre-Filling & Validation

DETERMINISTIC BRIDGE

Under the ICAI Gazette notification, all tax audit reports, certificates, and GST audits require an 18-digit **Unique Document Identification Number (UDIN)** generated within 60 days of signing. Failure to generate UDIN invalidates the certificate and triggers professional misconduct proceedings.

- **Pre-Calculated Figures:** Stratum 1 auto-compiles the exact 3 figures required by ICAI (Turnover, Net Profit, Assessed Value) directly from certified return working papers.
- **Direct Headless ICAI Portal Bridge:** Pre-fills the UDIN form, presenting the Partner with a 1-click verification screen and mobile OTP prompt.
- **WORM Ledger Locking:** Once the 18-digit UDIN is returned, it is permanently etched into the audit working paper and cannot be edited.

7.0 THE FOUR-EYES PRINCIPLE (MAKER-CHECKER INTEGRITY)

SQC 1 COMPLIANCE

1. The Preparer (Maker):
Article assistant or junior staff processes OCR extractions, clears exceptions, and prepares the draft GSTR-3B/ITR return. The preparer's cryptographic user ID is permanently bound to the draft.

2. Secondary Reviewer (Checker):
A separate qualified senior staff member or manager inspects variance logs, Rule 38 checks, and reconciliations. The system mathematically prevents the Maker from approving their own work.

3. Partner Authorization (Signer):
The Partner reviews the Four-Eyes Verification Certificate, confirms absence of blocking rule alerts, and executes Class 3 DSC token signing. The entire chain forms an un-alterable audit trail.

8.0 THE GST INDIRECT TAX OPERATING ENGINE

COMPLETE INDIRECT TAX WORKFLOW

GST in India has evolved into an interconnected, multi-system compliance maze. With the statutory rollout of the **Invoice Management System (IMS)** by GSTN, recipient taxpayers must actively triage supplier invoices prior to GSTR-2B compilation. KARYA provides an end-to-end, multi-GSTIN indirect tax pipeline engineered for Indian CA firms.

1. Outward Reconciliation (GS-1)

SALES ENGINE

Performs four-way reconciliation: **Sales Register (Books) vs GSTR-1 draft vs E-Way Bill Portal vs E-Invoice IRN repository**. Detects missing IRNs, cancelled e-way bills, and B2B vs B2C classification errors before upload.

2. Inward Triage & IMS (GS-2)

IMS ENGINE

Direct API sync with GSTN Invoice Management System. Inward supplier bills are automatically classified: **Accept (Eligible), Reject (Erroneous/Duplicate), or Pending (Awaiting Delivery/Verification)**.

3. Dynamic ITC Matching (GS-3)

CREDIT ENGINE

Matches Purchase Register against GSTR-2B. Automatically enforces **Section 16(2)(aa)** filing conditions, **Section 17(5)** blocked credits, and **Rule 37 180-day** non-payment reversal tracking.

9.0 DEEP DIVE: THE INVOICE MANAGEMENT SYSTEM (IMS) ENGINE

2024-2026 STATUTORY MANDATE

The Invoice Management System (IMS) fundamentally alters recipient credit rights. Inward invoices reported by suppliers on GSTR-1/1A flow to IMS. If a CA firm fails to triage these invoices, the system automatically treats them as accepted, potentially locking the client into ineligible credit or tax liabilities.

- **Automated Classification Matrix:** GS-2 evaluates inward invoices against client purchase orders and warehouse receipts. Invoices with GSTIN mismatches, wrong POS, or duplicate bill numbers are flagged for *Reject*.
- **The "Pending" Holding Area:** Invoices for goods in transit at month-end are tagged *Pending*, preventing premature credit claims while deferring them to subsequent tax periods without losing ITC rights.
- **Client Proposal Portal:** SME clients can review suggested IMS actions via a simplified mobile/web UI. However, per **Law 4**, client actions remain proposals until committed by firm staff at Gate G3.

IMS ACTION STATE MACHINE

Accept	Flows to GSTR-2B; Eligible for GSTR-3B ITC
Reject	Sent back to supplier; Zero ITC; Prevents audit demand
Pending	Carried forward; Allowed in future month upon delivery

Rule 88D Defense: Proper IMS triage eliminates GSTR-2B vs 3B mismatch intimations.

10.0 DRC INTIMATION DEFENSE: RULE 88C (DRC-01B) & RULE 88D (DRC-01C)

AUTOMATED TAX NOTICE MITIGATION

DRC-01B Containment: GSTR-1 vs GSTR-3B Liability (Rule 88C)

OUTWARD DEFENSE

Under Rule 88C, if outward tax liability reported in GSTR-1 exceeds liability paid in GSTR-3B by more than a specified percentage/amount, the GSTN portal issues an automated DRC-01B intimation. If unpaid or unexplained within 7 days, subsequent GSTR-1 filing is hard-blocked.

KARYA Pre-Filing Barrier: GS-5 executes hard-rule check #12 prior to GSTR-3B lock. If GSTR-1 exceeds 3B, it halts execution, displays the exact delta schedule, and forces the user to link a DRC-03 voluntary payment or document the statutory explanation.

DRC-01C Containment: GSTR-2B vs GSTR-3B Credit (Rule 88D)

INPUT CREDIT DEFENSE

Under Rule 88D, claiming ITC in GSTR-3B exceeding the credit available in GSTR-2B triggers an automated DRC-01C mismatch demand with a 7-day portal lock penalty.

KARYA Pre-Filing Barrier: Stratum 1 applies a hard ceiling on Table 4(A)(5) all other ITC. Credit claimed cannot exceed 2B without an explicit, partner-certified variance note explaining spillover credit or reversal cancellations under Rule 38.

11.0 DETERMINISTIC TAX MATH: SECTION 50 NET CASH INTEREST & SUPPLIER RECOVERY

PRECISION FINANCIAL ENGINE

Section 50 Interest Engine (Dated Python Code)

RUST/PYTHON ENGINE

Per the proviso to Section 50(1), interest on delayed filing is payable strictly on that portion of the tax paid by debiting the **Electronic Cash Ledger**, never on the gross liability offset via credit ledger. KARYA's dated math module computes rupee-accurate interest day-by-day, preventing overpayment of interest to the exchequer while defending the client against portal calculation bugs.

GS-4: Autonomous Supplier Recovery Notice Engine

CASH RECOVERY

When suppliers fail to file GSTR-1, the client's ITC is lost under Section 16(2)(c). GS-4 detects non-filing vendors from GSTR-2A vs 2B diffs, drafts customized legal recovery notices citing invoice dates and GSTINs, and sends them via email/WhatsApp, recovering lakhs in blocked capital for SME clients.

12.0 DIRECT TAX & WITHHOLDING (TDS/TCS) ARCHITECTURE

INCOME TAX ACT COMPLIANCE

Direct tax compliance for Indian businesses demands strict adherence to quarterly TDS returns, annual ITR filings, and cross-verification against government big-data intelligence systems. KARYA automates the painful clerical burden of tax deduction classification, AIS reconciliation, and advance tax estimation.

DT-1: TDS/TCS Withholding & FVU Generation

TDS COMPLIANCE

Withholding tax non-compliance attracts devastating 30% business expense disallowances under Section 40(a)(ia) and penal interest under Section 201(1A).

- **Autonomous Section Disambiguation:** Classifies contracts into correct statutory buckets: **194C (Works Contract - 1%/2%) vs 194J(a) (Technical Services - 2%) vs 194J(b) (Professional Services - 10%)**.
- **Section 194Q vs 206C(1H) Buyer/Seller Hierarchy:** Deterministically applies threshold tests (>₹50 Lakhs turnover + >₹50 Lakhs purchase) to eliminate double deduction or non-deduction.
- **NSDL FVU Bridge:** Generates quarterly Form 24Q (Salary) and Form 26Q (Non-Salary) e-TDS files, executing the official NSDL File Validation Utility (FVU) locally.

DT-2: AIS, TIS & Form 26AS Three-Way Reconciler

CROSS-CHECK ENGINE

The Income Tax Department's Annual Information Statement (AIS) and Tax Information Summary (TIS) aggregate data from banks, registrars, mutual funds, and GSTN. Mismatches trigger automated Section 143(1)(a) adjustments.

- **Automated Ingestion:** Ingests JSON/PDF exports of 26AS, AIS, and TIS directly from the Income Tax portal.
- **Three-Way Ledger Match:** Reconciles reported receipts against client turnover and checks TDS credit against balance sheet advance tax.
- **Uncredited TDS Recovery:** Flags deductions where vendors deducted tax but failed to deposit it or entered incorrect PANs, preventing tax credit leakage.

13.0 SECTION 43B(H) MSME 45-DAY PAYMENT COMPLIANCE ENGINE

CRITICAL AUDIT & TAX RISK

The introduction of **Section 43B(h) of the Income Tax Act** requires any sum payable to a **Micro or Small Enterprise** beyond the time limit specified in Section 15 of the MSMED Act, 2006 (15 days without agreement, maximum 45 days with written agreement) to be **disallowed and added back to taxable income** for the year. This represents one of the largest tax audit exposures for CAs and SME clients today.

- **Automated Udyam Portal Verification:** DT-1 / AU-2 automatically validates supplier Udyam Registration numbers via API, verifying enterprise category (Micro, Small, or Medium). Medium enterprises are exempted from Section 43B(h).
- **Ageing Analysis & Written Agreement Matrix:** Calculates exact credit days from invoice date / acceptance date. Segregates payments made within 15/45 days vs overdue balances as of March 31st.
- **Form 3CD Clause 22 Auto-Population:** Auto-computes interest due under Section 16 of the MSMED Act (compounded monthly at three times RBI Bank Rate) and populates Clause 22 disallowance schedules with zero manual calculation.

SECTION 43B(H) STATUTORY RULES

Micro Enterprise	Investment ≤ ₹1 Cr & Turnover ≤ ₹5 Cr
Small Enterprise	Investment ≤ ₹10 Cr & Turnover ≤ ₹50 Cr
Medium Enterprise	EXEMPT from Section 43B(h) disallowance
Max Credit Period	15 Days (Default) / 45 Days (Written Agreement)
Tax Consequence	100% Disallowed in P&L; Taxed at 25-30%

14.0 ADVANCE TAX PROJECTION & ITR PRE-FILLING (DT-3 & DT-4)

STATUTORY SCHEDULE COMPILATION

DT-3: Advance Tax Quarterly Projection Engine

CASH MANAGEMENT

Indian taxpayers must remit advance tax in four statutory installments: 15% by June 15, 45% by Sept 15, 75% by Dec 15, and 100% by March 15. Shortfalls trigger mandatory 1% monthly penal interest under Section 234B and 234C.

Continuous Forecast Engine: Extrapolates annualized net profit from live Tally/Busy ledgers, deducts available TDS credits from 26AS, factors in Section 43B(h) disallowances, and prepares quarterly challan drafts with exact rupee amounts.

DT-4: ITR-1 to 7 Schedule Compilation

SCHEMA GENERATION

Indian Income Tax Return schemas require complex multi-schedule cross-population (Schedule DAA, Schedule BP, Schedule CYLA, Schedule BFLA, MAT/AMT schedules).

Deterministic JSON Generation: Maps audited financial statements into valid CBDT JSON schemas. Performs automated tax calculation under both Old and New Tax Regimes (Section 115BAC), presenting a comparative benefit sheet to the Partner at Gate G14.

15.0 TAX AUDIT FORM 3CD CLAUSE GENERATOR (AU-2)

ICAI TAX AUDIT COMPLIANCE

Tax Audit under Section 44AB requires reporting across 44 exhaustive clauses in **Form 3CD**. A single misstatement or missing reporting item can result in disciplinary proceedings before the ICAI Board of Discipline or penalty under Section 271B. AU-2 automates data extraction across all 44 clauses while preserving partner sign-off.

FORM 3CD CLAUSE	STATUTORY DESCRIPTION	AUTOMATED EXTRACTION & VERIFICATION LOGIC	AUDIT EVIDENCE GENERATED
Clause 13	Method of Accounting & ICDS	Validates consistency with previous year; checks compliance with 10 Income Computation & Disclosure Standards (ICDS).	Accounting policy memo & ICDS deviation schedule.
Clause 17	Land / Building Transfer (Sec 43CA / 50C)	Compares consideration against stamp duty valuation from registered conveyance deeds.	Stamp valuation vs agreement variance workpaper.
Clause 21(a)	Capital / Personal / Penalty Disallowance	Parses ledger narration for fines, penalties, compounding fees, and personal drawings.	Itemized penalty extract with ledger voucher numbers.
Clause 21(d)	Cash Payments >₹10,000 (Sec 40A(3))	Scans cash payment vouchers across all bank and cash ledgers for payments exceeding statutory threshold.	Rule 6DD exemption verification sheet.
Clause 22	MSME Interest Dues (Sec 23 MSMED)	Automated calculation of compound interest due to micro/small vendors overdue under Section 43B(h).	Supplier-wise interest computation register.
Clause 26	Section 43B Statutory Dues	Verifies timely deposit of GST, PF, ESIC, bonus, and leave encashment prior to ITR due date.	Challan deposit date vs due date matrix.
Clause 34	TDS / TCS Compliance & Payables	Cross-verifies chapter XVII-B deductions, late deductions, and non-deductions against quarterly 26Q/24Q returns.	TDS ledger reconciliation vs Form 26AS.
Clause 44	Breakup of Total Expenditure in GST	Automated categorization of total expenditure: registered entities, exempt goods, composite dealers, and unregistered entities.	GSTR-2B vs P&L expenditure allocation table.

16.0 AUDIT WORKING PAPERS & SA 230 STANDARDS ENGINE (AU-1)

SQC 1 AUDIT TRAIL

ICAI Standards on Auditing (SA 230) Compliance

DIGITAL AUDIT FILE

SA 230 (Audit Documentation) requires auditors to prepare working papers that provide sufficient and appropriate record of the basis for the auditor's report. Incomplete working papers represent the #1 cause of adverse ICAI Peer Review findings.

- **Standardized Electronic Working Papers:** AU-1 auto-indexes trial balances, lead schedules, substantive test sheets, and balance confirmations into standardized ICAI paper templates.
- **Monetary Unit Sampling (MUS):** Implements statistical sampling for debtors, creditors, and inventory verification based on firm-defined materiality thresholds.
- **Audit Query Tracking:** Logs article assistant audit queries to client management with timestamped resolution histories and attachment proofs.

External Balance Confirmation Automation

SA 505 COMPLIANCE

SA 505 governs external confirmation procedures for trade receivables and bank balances.

- **Automated Ledger Extraction:** Extracts top debtors and creditors per sampling rules.
- **Secure Direct Email / WhatsApp Bridge:** Generates digitally signed confirmation requests sent directly to vendor/customer financial contacts.
- **Reconciliation Ledger:** Ingests counterparty statements and auto-generates debtor/creditor reconciliation statements.

17.0 SECTION 141(3)(G) 60-AUDIT STATUTORY CEILING TRACKER (FO-1)

PARTNER STATUTORY PROTECTION

Under **Section 141(3)(g) of the Companies Act, 2013**, a Chartered Accountant is prohibited from accepting statutory audit appointments for more than **20 companies** (excluding one-person companies and small companies with paid-up capital <₹100 Cr). Furthermore, ICAI guidelines impose an aggregate signing ceiling of **60 statutory company audits per partner annually** across all firms in which they are a partner.

- **Cross-Firm Global Signing Ledger:** FO-1 maintains a real-time, tamper-proof count of all audit appointments signed by each partner across all associated partnerships and LLPs.
- **Hard Pre-Appointment Blocking:** If an engagement would cause a partner to breach the statutory limit of 60 audits, the system hard-blocks appointment acceptance at Gate G13.
- **Capacity Forecast & Reallocation Engine:** Evaluates ongoing audits in April and forecasts peak season loads, allowing managing partners to safely reallocate corporate audits across partners before July deadlines.

60

MAX AUDITS / PARTNER / YEAR

Companies Act Sec 141(3)(g) + ICAI Ethical Standards Ceiling

SYSTEM HARD-BLOCK

18.0 TALLYPRIME & BUSY BI-DIRECTIONAL ERP SYNCHRONIZER

SEAMLESS ACCOUNTING WORKFLOW

Over 85% of Indian SMEs and CA firms rely on **TallyPrime** or **Busy Accounting Software**. Any platform that requires clients to migrate to a new accounting software is rejected by Indian practice. KARYA operates as an intelligent bi-directional sync layer directly atop the client's existing desktop TallyPrime or Busy installations.

Journal Entry Voucher (JEV) XML Sync Protocol

TALLY XML PROTOCOL

KARYA communicates with TallyPrime via local XML-over-HTTP port 9000 or ODBC without requiring cloud migration or database tampering.

- **Direct Voucher Posting:** Validated purchase bills, bank statements, and TDS entries are posted directly into Tally as native `VOUCHER` XML entities.
- **Chart of Accounts Introspection:** Auto-fetches client ledgers, stock items, and tax heads. Adapts to idiosyncratic ledger naming (e.g. "Input CGST @ 9% (Local)" vs "Central Tax Input").
- **Back-Dated Edit Detection:** Continuously monitors Tally altered voucher tables; flags any back-dated modification in locked compliance periods to prevent post-audit variance.

Multi-Bank Statement Parser & Narration NLP (IN-4)

BANK RECONCILIATION

Bank reconciliation is traditionally the most time-consuming clerical bottleneck in Indian accounting.

- **30+ Indian Bank Parsers:** Native format parsers for HDFC, ICICI, SBI, Axis, Kotak, PNB, and cooperative banks, handling un-delimited PDFs and password-protected statements.
- **Narration Mining:** Decodes messy string patterns (e.g. `NEFT-INW-YESB000123-ABC-CORP-GSTIN27AA...`) to automatically extract counterparty GSTINs, PANs, and vendor names.
- **Auto-Contra Matching:** Identifies inter-bank transfers and petty cash withdrawals, matching debit/credit legs across multiple client accounts.

19.0 WHATSAPP BUSINESS API (WABA) CLIENT CONCIERGE & CHASE LADDER

AUTOMATED DOCUMENT HARVESTING

Indian business owners manage their operations via WhatsApp. Email requests from CA firms go unanswered for weeks. **IN-2 executes an enterprise WhatsApp Business API (WABA) conversational chase engine** engineered with polite, culturally calibrated Indian professional etiquette.

- **Automated Progressive Chase Ladder:** Nudges clients for missing purchase bills and bank statements on a structured cadence: **3 days prior to due date, 1 day prior, and morning of due date**.
- **Multimodal Ingestion:** Clients can simply photograph purchase invoices or forward PDF bank statements directly to the firm's verified WhatsApp number.
- **Automated Password Decryption:** Statements encrypted with standard Indian passwords (PAN + Date of Birth / Customer ID) are automatically decrypted in quarantine memory.

Client WhatsApp Interaction Flow

*"Dear Sharma Ji, kindly share September bank statement for HDFC A/c **4391 to finalize GSTR-3B."*

"Attached HDFC_Sept.pdf (Pass: SHAR0805)"

"Received! 142 transactions parsed and reconciled with Tally."

20.0 FRONT DESK GUARD (CL-1) & TAX NOTICE PARSER (CL-2)

DEFENSIVE CLIENT INTERACTION

CL-1: The Closed 7-Question Front Desk Guard

ETHICAL BARRIER

Unconstrained AI chat interfaces are disastrous in tax practice. A seemingly innocent question like "*Can I claim ITC on my car purchase?*" requires legal interpretation of Section 17(5)(a). An AI that improvises creates catastrophic professional liability.

The Closed Whitelist Architecture: CL-1 is restricted strictly to a closed whitelist of **seven factual operational questions**: (1) Has my return been filed? (2) What documents are outstanding? (3) When is my next statutory due date? (4) Send my filing acknowledgement. (5) Did you receive my documents? (6) How many IMS bills await my action? (7) Who is my assigned accountant?

Gate G10 Default Refusal: Any legal, tax planning, notice interpretation, or fee dispute question is immediately halted and routed to the assigned staff member. Refusal is the default.

CL-2: Statutory Tax Notice & Demand Parser

NOTICE RESPONSE PIPELINE

SME clients frequently panic upon receiving automated government intimations and forward unread notice PDFs.

- **Instant Multi-Section Classification:** Parses GST notices (DRC-01, DRC-01A, ASMT-10) and Income Tax notices (Section 143(1), Section 148 reassessment, Section 139(9) defective return).
- **Demand & Limitation Extractor:** Extracts demand tax, interest, penalty, and exact limitation reply deadline (auto-creating an emergency job with a 5-day safety buffer).
- **Gate G11 Partner Reservation:** The agent drafts zero words of legal rebuttal. The statutory reply must be drafted and signed exclusively by the licensed Partner.

21.0 PRACTICE ECONOMICS & BILLING REALIZATION (FO-3)

COMMERCIAL PRACTICE TRANSFORMATION

Most CA firms suffer from massive revenue leakage: **out-of-scope work** (drafting notice replies, answering scrutiny queries, reconciling messy prior-year books) is performed by staff but never billed to clients due to lack of time tracking. FO-3 transforms the firm's commercial economics.

PRACTICE METRIC	TRADITIONAL CA PRACTICE	WITH KARYA OPERATING SYSTEM
Turnaround Time (GSTR-3B)	12 to 16 days (chasing + manual entry)	**48 hours** (automated extraction & triage)
Capacity per Article Assistant	15 to 25 SME clients max	**60 to 80 SME clients** (3.5x leverage)
Out-of-Scope Revenue Recovery	<15% captured; mostly unbilled	**85%+ captured** via automated FO-3 billing logs
DRC-01B / 01C Penalty Exposure	Frequent month-end panic & interest	**Zero penalty incidents** (hard-rule barriers)
Partner Billable Realization	40% lost to administrative review	**82% billable time** on advisory & structuring

+42%

NET FIRM REALIZATION

Across pilot deployments with 5-partner firms handling 300+ clients, billable realization increased by 42% in Year 1.

22.0 ADVERSARIAL QUALITY CONTROL & SQC 1 PEER REVIEW (FO-5)

QUALITY CONTROL ARCHITECTURE

Adversarial Canary Testing (Preventing "Rubber-Stamping")

QUALITY DEFENSE

The greatest operational risk in automated tax software is that tired article assistants stop critically reviewing computer proposals and click "Approve" blindly (rubber-stamping).

Synthetic Canary Injection: FO-5 intermittently injects known synthetic errors (e.g. an ineligible personal vehicle expense or an expired GSTIN) into the reviewer's queue. If the reviewer approves the defective entry without inspection, the incident is flagged to the Quality Partner.

ICAI Peer Review Readiness & Immutable Audit Vault

SQC 1 AUDIT TRAIL

Under the ICAI Peer Review Mandate, firms auditing listed and unlisted public companies must undergo rigorous peer review audits of their quality control mechanisms (SQC 1).

One-Click Peer Review Package: The platform produces an immutable, cryptographic WORM ledger recording every user action, DSC signature hash, document source, and rule check version, satisfying ICAI Peer Reviewers instantly.

23.0 DATA SOVEREIGNTY, PRIVACY & ENTERPRISE SECURITY ARCHITECTURE

GOVERNMENT-GRADE SECURITY

1. Sovereign Data Residency

All client databases, documents, and backups are strictly hosted in Tier-4 data centers within the **Republic of India** (AWS Mumbai / Azure Pune). Zero offshore transmission.

2. Zero Data Retention (ZDR)

All multimodal OCR and language models operate under signed **Zero Data Retention (ZDR)** agreements. Client financial records are never retained or utilized to train underlying models.

3. PostgreSQL Tenant Isolation (RLS)

Enterprise multi-tenancy enforced via **PostgreSQL Row-Level Security (RLS)**. Client records are cryptographically partitioned, preventing cross-client data leakage even across staff accounts.

24.0 ICAI DISCIPLINARY & LEGAL SHIELD: PROTECTING THE SIGNING PARTNER

PERSONAL STATUTORY LIABILITY

Gross Negligence Defense (CA Act)

ETHICAL DEFENSE

Protects against Clause (7), Part I, Second Schedule (gross negligence) by maintaining a cryptographic Four-Eyes review certificate (Gate G6) and dated rule engine logs for every signed figure.

Confidentiality Protection

CLIENT SECRECY

Complies with Clause (1), Part I, Second Schedule (non-disclosure of client information) via Indian sovereign VPC hosting, Zero Data Retention (ZDR), and zero public AI model training.

Criminal Prosecution Shield

STATUTORY SHIELD

Prevents liability under GST Section 132 / IT Act Section 277 via Law 1 (zero autonomous filing), ephemeral Class 3 DSC token RAM (<30s), and hard-blocking pre-filing checks (DRC-01B/01C).

KARYA: BUILT EXCLUSIVELY FOR THE INDIAN CA PROFESSION

Preserving partner liability, defending client capital, and elevating the Chartered Accountant from clerk to strategic counsel.

ICAI SQC 1 Compliant Architecture

Version 4.2.0 • Enterprise Edition